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A-A-53974
11 September 1989

COMMERCIAL ITEM DESCRIPTION

BUR KIT, DENTAL FINISHING (Tungsten Carbide)

The General Services Administration has authorized the use of this Commercial Item Description.

This commercial item description covers a dental tungsten carbide finishing bur kit for use with friction grip handpieces.

Salient characteristics:

Head sizes. The bur kit shall consist of the following sizes:

ET3	ET9
ET4	OS1
ET6	OS2

Material. The burs shall be of two piece construction. The head of the burs shall be fabricated from a solid piece of cemented tungsten carbide with cobalt binder. The shank of the burs shall be fabricated from martensitic corrosion resisting steel type 420 in accordance with ASTM F899 with a carbon content of 0.4 to 0.5 percent and a maximum phosphorus content of 0.45 percent. As an alternate the shank shall be fabricated from nickel or chromium plated AISI-S5 tool steel. The bur block shall be fabricated from acrylic plastic.

Hardness. The bur head shall have a Vickers hardness of 1600 (minimum). The bur shank shall have a Vickers hardness of 300 (minimum). The bur shall be tested for hardness in accordance with ASTM E92. A 3 kgf test load shall be used for the hardness testing.

AMSC N/A FSC 6520

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Style, and design. The burs shall be designed for use in trimming and finishing composite resin and acrylic restorations, beveling margins in crown, inlay and onlay preparations, cosmetically contouring teeth and removing gold or amalgam overhangs. The burs shall be designed for use with friction grip handpieces. The burs shall consist of a shank, a neck and a head. The blades shall be uniformly cut and sharpened. The burs head shall be permanently welded to the shank. The kit shall include one bur block and 2 burs each of the following sizes: ET3, ET4, ET6, ET9, OS1 and OS2. The style and design of the bur shank shall be in accordance with Figure 1. The style and design of the bur sizes ET3, ET4, ET6 and ET9 shall be in accordance with Figure 2. Bur size OS1 shall be in accordance with Figure 3. Bur size OS2 shall be in accordance with Figure 4. The values listed below refer to figures 1 to 4.

Bur Size	C (Diam) mm ± 0.03	D (Diam) mm ± 0.05	E (Total both sides) degrees	F (Diam) mm ± 0.05	Overall length mm ± 0.5
ET3	0.23	0.68	9-12	0.80	23.0
ET4	0.23	0.88	9-12	1.00	20.0
ET6	0.23	1.10	9-13	1.40	21.0
ET9	0.23	1.10	6-8	1.40	24.0
OS1	2.3	1.28	----	----	19.0
OS2	0.6	0.78	5-7	0.90	20.0

Bur Size	Head Diam 0.5 mm from tip mm ± 0.03	KL mm ± 0.1	B degrees	F degrees	Number of Blades
ET3	0.33	3.0	65-70	18-22	8
ET4	0.33	4.2	65-70	18-22	8
ET6	0.33	6.0	65-70	18-22	8
ET9	0.30	9.0	65-70	18-20	8
OS1	----	4.2	85-95	28-32	12
OS2	----	3.2	85-95	28-32	12

Bur block. An acrylic bur shall be provided. The bur block shall be designed to hold the 12 burs. The bur block shall have permanent markings to identify each different size bur and shall have a transparent cover.

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Runout. The total indicated runout shall not exceed 0.05 mm when tested as specified herein.

Strength of neck and joint. The neck shall not break and the bur shall not fracture when tested as specified herein. The nondamaging test load shall be as specified below:

Bur Size	Nondamaging test load, newtons
ET3	6
ET4	10
ET6	12
ET9	10
OS1	14
OS2	8

Increase in runout. The increase in runout shall be no greater than 0.05 mm when tested as specified herein on the samples subject to the strength of neck and joint test.

Finish. The bur head shall have a finely honed finish with sharpened blades.

Autoclaving. The bur shall show no corrosion when tested as specified herein, except that any corrosion present at the welded joint shall be disregarded and shall not be cause for rejection.

Chemical disinfection and sterilization. The bur shall be capable of undergoing chemical disinfection, dry heat chemiclave and hot oil without showing any signs of corrosion.

Surface roughness. The surface roughness of the bur shank shall not exceed a center line average (CLA) value of 0.5 micrometers when tested as specified herein.

Identification marking. Marking of the bur with the name or trademark shall be at the option of the supplier.

Workmanship. The bur shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One bur kit containing two each of six friction grip burs, and one bur block, as specified, constitutes one unit.

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Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, and the lot size.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, an inspection level of II. Examination shall include, but shall not be limited to the defects listed below. The unit of product for examination purposes shall be one bar of each size.

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Defects

Head shape not as specified.
Bur sizes not as specified.
Number of blades not as specified.
Laminations, cracks, bends, deformed portions, gouges, chips or broken areas present in any portion of the bur.

Microscopic inspection. Examination of bur defects not readily perceptible to the naked eye shall be made using a microscope at 10 power (10X) magnification.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105. The unit of product for test purposes shall be one bur of each size. Sufficient amount from each lot of each metal used in the construction of the bur shall be selected for chemical analysis with an acceptance number of zero for all sample sizes. Sampling for dimensions shall be conducted at inspection level S-2. Sampling for all other tests shall be conducted at inspection level S-1.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Chemical composition. In the absence of certificates of quality for the raw steel used in the fabrication of the bur mandrels, chemical analysis of the mandrels shall be determined in accordance with Method 111 or 112 of FED-STD-151. In case of a dispute, method 111 shall be used for referee analysis.

Dimensions. Inspection of head and shank diameters shall be accomplished with regularly calibrated tungsten carbide ring gauges, toolmaker's microscope, or other calibrated instruments of equal accuracy. Neck diameters shall be inspected with a calibrated blade-type micrometer caliper or equivalent. Fitting dimensions shall be checked with appropriate gauges with tungsten carbide faces. Overall length and dimensions shall be checked with a standard tungsten carbide faced micrometer caliper or with a toolmaker's microscope or equivalent. A toolmaker's microscope or an optical comparator shall be used for measuring cone angles. For a true reading of the cone angles, the reading must be made on one blade on one side of the bur head, reading the angle by partial rotation of the bur, equal to the spiral of the blade, in a V block. The angle so measured shall be doubled to give the cone angle.

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Runout. The bur shall be placed and held in a "V" block or precision chuck, and rotated slowly. A dial indicator sensitive to 0.005 mm and equipped with a knife-edge foot, or a shadowgraph or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, the indicator shall be mounted with the shaft perpendicular to the axis of the bur, and so positioned that the foot contacts the bur neck at the point of smallest diameter. The maximum and minimum readings obtained during one complete 360-degree rotation of the bur shall be measured and recorded to the nearest 0.01 mm. The difference between these two readings shall be taken as the total indicated runout.

Neck and joint strength. The testing equipment shall consist of a true-running chuck suitable for holding and rotating the test bur, and a loading device capable of applying the load through an anvil with a flat surface at an angle of 22-1/2 ± 1/2 degrees to the bur axis. The chuck shall grip at least a 9.5 mm portion of the bur shank starting at the intersection of the neck and shank.

Following an initial examination for total indicated runout, those burs which have been selected for the neck and joint strength test shall be rotated in the chuck to that position which shows the highest deflection reading on the dial indicator gage. A load of the magnitude specified herein shall be slowly applied at an angle to the bur axis of 22-1/2 ± 1/2 degrees, in the direction which tends to further increase the dial deflection reading. The neck and joint of the bur shall be observed for evidence of breakage. Should breakage not have occured at the point of maximum deflection, the load shall be removed and the examination for increase in runout determined.

Increase in runout. The increase in runout shall be determined only on those samples which have been subjected to the neck and joint strength, and shall be performed in the manner already specified for runout. The difference between the final and the initial indicated runout shall be taken as the increase in runout.

Autoclaving. The test burs shall be cleaned in accordance with the instructions supplied by the manufacturer. The burs shall then be autoclaved at 132 ± 3 degrees centigrade (C) and 190 ± 7 kilopascal pressure for a minimum of 20 minutes. After the burs are removed from the autoclave, the entire surface area of the non-carbide portion of the bur shall be examined for evidence of corrosion.

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Surface roughness. The surface roughness of the shank shall be determined using a Brush Surfindicator or equipment of demonstrated equal quality. The surface roughness shall be determined in the longitudinal direction on the straight portion of the shank. A cut-off of 0.8 mm and a sweep length of 5 mm shall be used.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

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NOTE: The following National Stock Number is covered by this document:

National Stock Number	Item Identification
6520-01-264-0812	Bur Kit, Dental Finishing

Ordering data. Intermediate/exterior package quantities, labeling and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

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Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

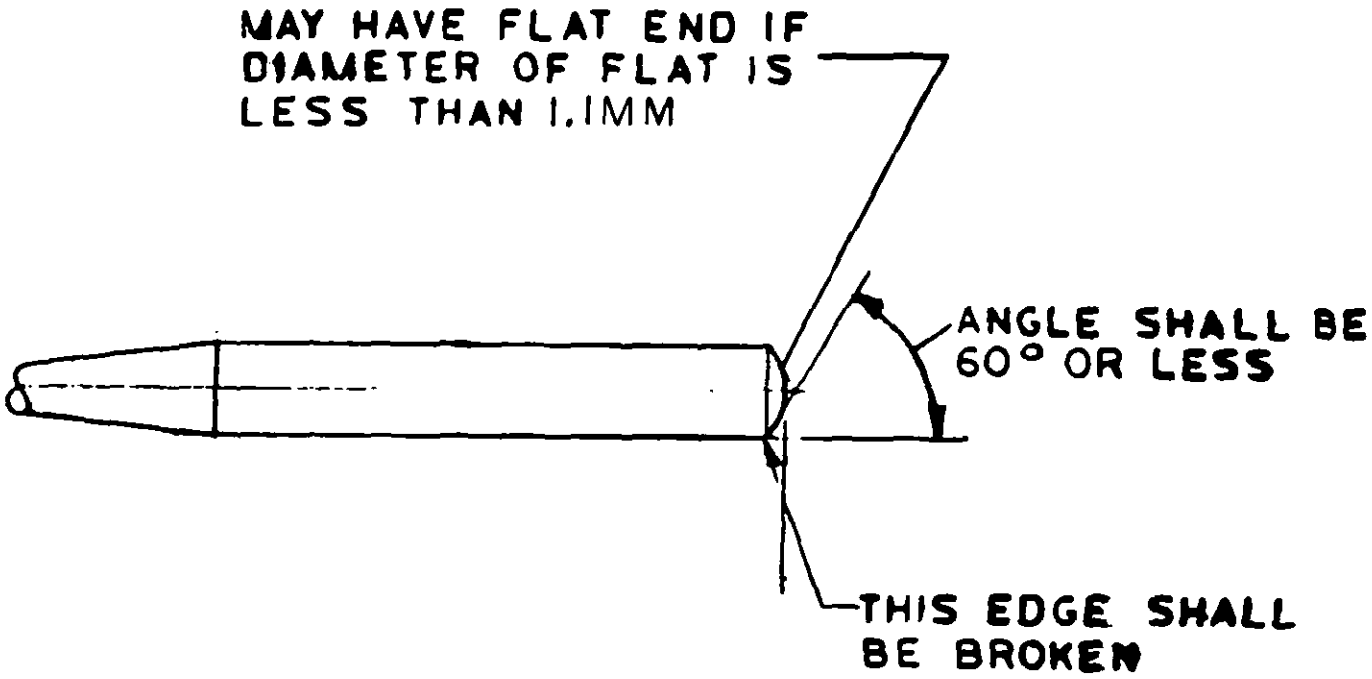
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The shank diameter shall be 1.590 to 1.600 mm

FIGURE 1

BUR SHANK

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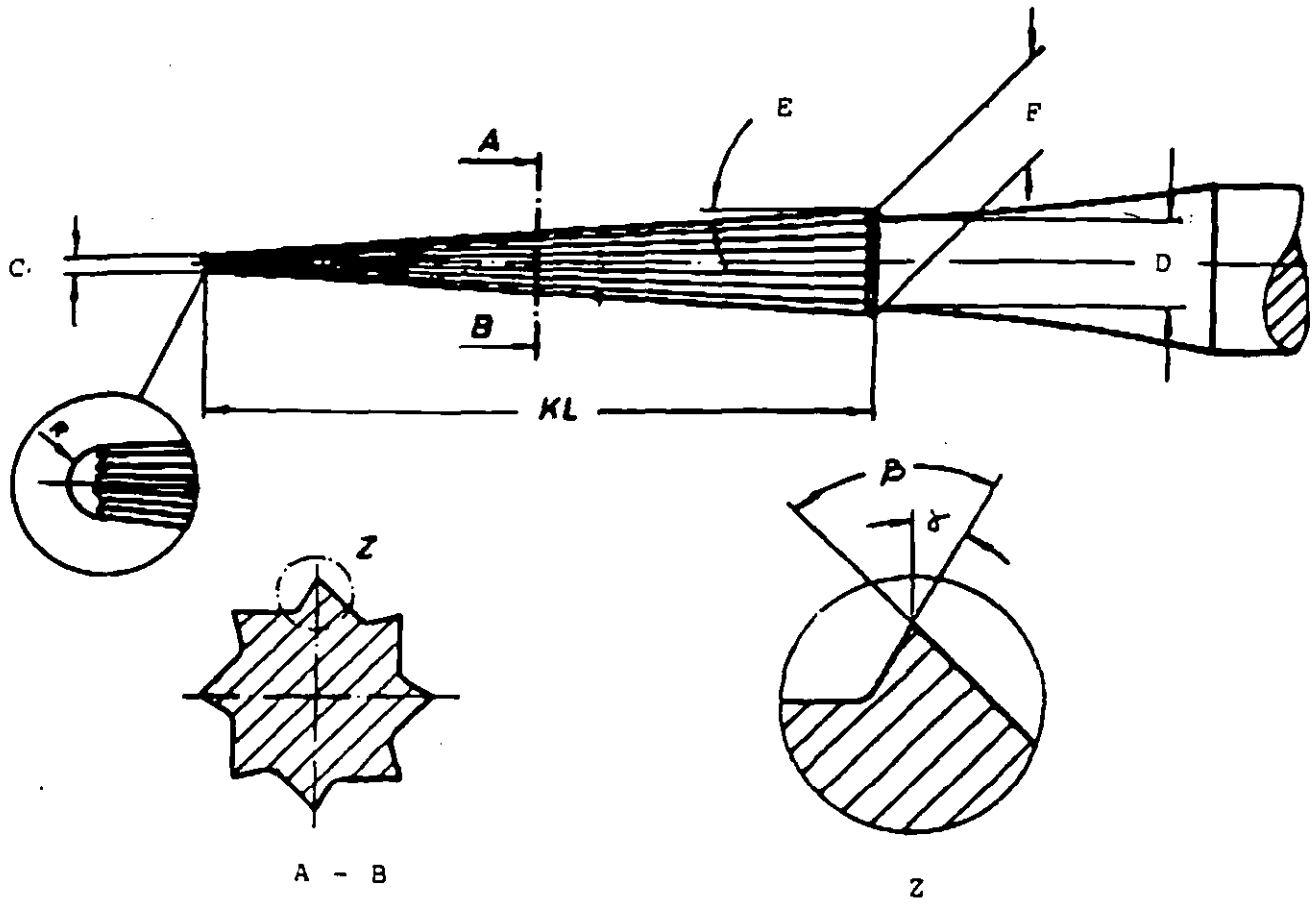


FIGURE 2
'E T' BURS

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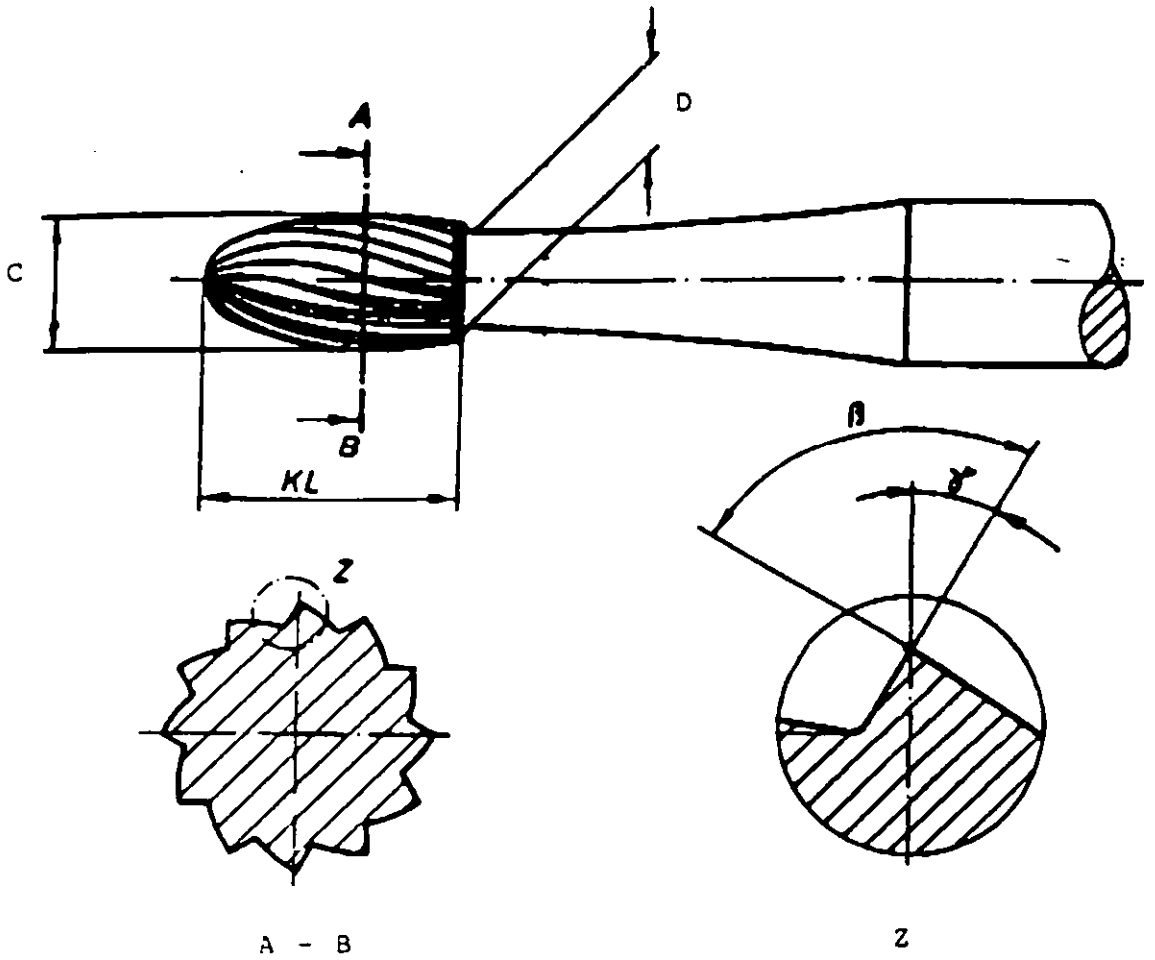


FIGURE 3
BUR SIZE OS1

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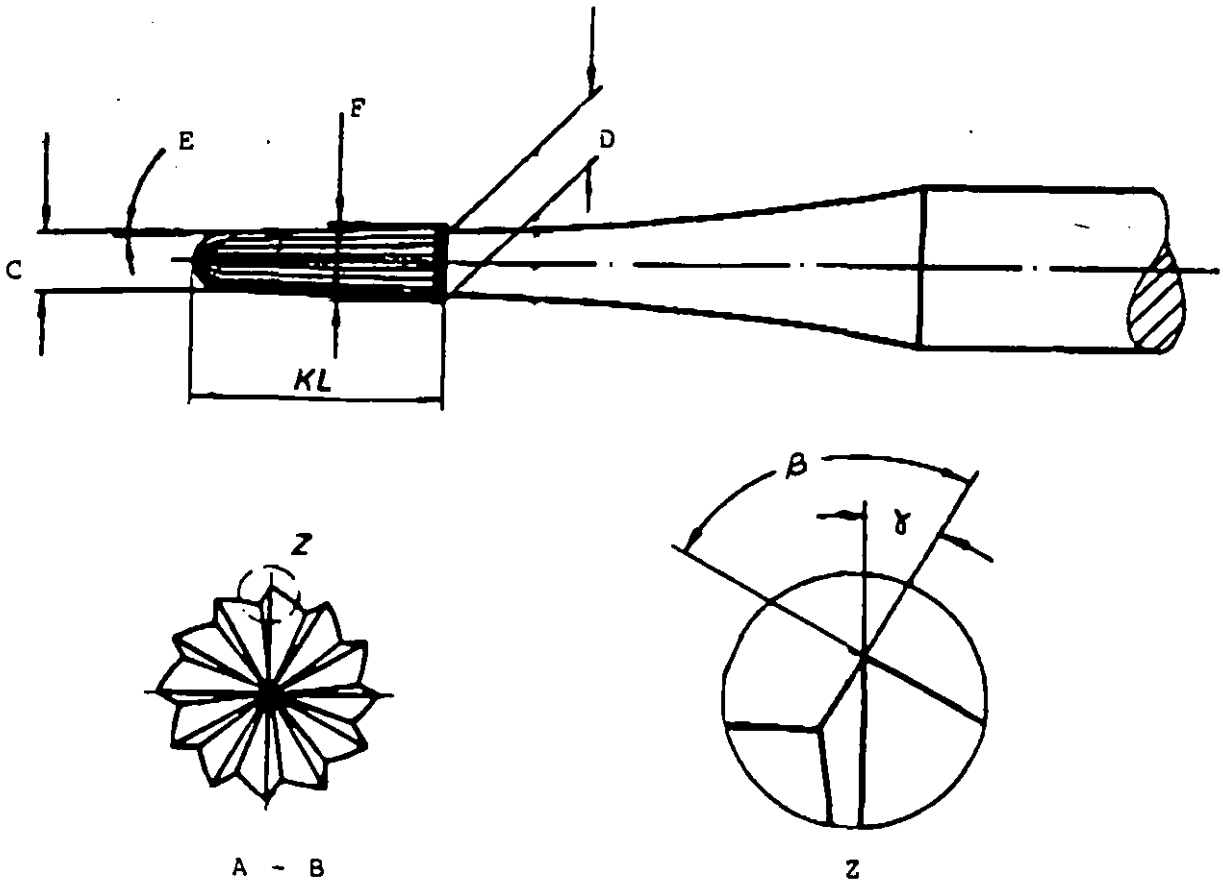


FIGURE 4
BUR SIZE OS 2

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-53974	BUR KIT, DENTAL FINISHING (Tungsten Carbide)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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